

Problem 1 (30 marks in total)

This problem must be done by hand without the aid of a computational device like a calculator or computer software like MATLAB

- (a) (10 marks) Determine whether the 3×3 matrix

$$\begin{bmatrix} 4 & 0 & 0 \\ -3 & 1 & 5 \\ 0 & 0 & 4 \end{bmatrix}$$

can be diagonalised or not.

- (b) (20 marks) Consider the quadratic form $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ given by

$$f(x) = 2x_1^2 - 8x_1x_2 + 2x_2^2 + 6x_2x_3 + 2x_3^2,$$

where $x = (x_1, x_2, x_3)^T$. Determine an orthogonal matrix V and real constants d_1, d_2, d_3 such that $f(Vz) = d_1z_1^2 + d_2z_2^2 + d_3z_3^2$, where $z = (z_1, z_2, z_3)^T$.

Problem 2 (40 marks in total)

- (a) (15 marks) Consider a system described by the ODE:

$$\begin{aligned} \dot{x} &= -x - 3y^5 \\ \dot{y} &= 2x^3 - y. \end{aligned}$$

Using the candidate Lyapunov function

$$V(x, y) = x^4 + y^6,$$

show that the origin is globally asymptotically stable.

- (b) Consider a control system described by the nonlinear controlled ODE:

$$\begin{aligned} \dot{x} &= -y(x - y) + u \\ \dot{y} &= -y(\sin x + x - y) + u, \end{aligned}$$

where $u = \eta(x, y)$ is a scalar state-feedback control signal. Answer the following questions:

- (i) (20 marks) Using the candidate Lyapunov function

$$V(x, y) = 1 - \cos x + \frac{1}{2}(x - y)^2,$$

show that you can design u such that it renders the origin asymptotically stable and $\dot{V}(x) = -\kappa \sin^2 x$ for some constant $\kappa > 0$.

- (ii) (5 marks) Using the same candidate Lyapunov function as in (i), is it possible to establish that the same control law u renders the origin globally asymptotically stable? Explain your answer.

Problem 3 (30 marks in total)

An autonomous linear time-invariant system (with no driving input) is described by the differential equation

$$\ddot{y} + 3\dot{y} + 2y = 0,$$

with initial conditions $y(0) = 1$ and $\dot{y}(0) = -2$. Using the observer canonical state-space representation for the system (with zero input), solve the state-space equation to find the solution of the differential equation *satisfying the given initial conditions* for all $t \geq 0$.